MIL-C-44009A
28 January 1982
SUPERSEDING
MIL-C-44009(GL)
26 July 1979

MILITARY SPECIFICATION

CHARGER, BATTERY, PP-7562/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a constant current battery charger designed for use with BB 541/u batteries.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
QQ-S-781	-	Strapping, Steel and Seals
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-P-291	-	Paperboard, Wrapping and Cushioning
PPP-S-760	-	Strapping, Nonmetallic (and Connectors)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6130

Source: https://assist.dla.mil -- Downloaded: 20260-21T23:21Z
Check the source to verify that this is the current version before use.

PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

8-1-375 - Cable Assembly, Constant Current Battery Charger, Modular Battery Pack

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

UNDERWRITERS LABORATORIES, INC.

STD. NO. 1236 - Battery Chargers, Electric

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:21Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Standard product. The battery chargers delivered under this specification shall be the manufacturer's current standard product except for any changes necessary to meet the requirements of this document.

3.2 Codes and standards. The battery chargers shall comply with the requirements of UL Standard No. 1236. Prior to approval of the first shipment the contractor shall submit satisfactory evidence to the contracting officer or his representative that the battery chargers he proposes to furnish under this specification meet the requirements of UL Standard No. 1236. Evidence of this shall be the UL listing mark, or a certified test report from a recognized independent testing laboratory acceptable to the government indicating this compliance.

3.3 Construction. The size of the battery charger shall not exceed 8.5 inches x 5.2 inches x 2.6 inches and its weight shall not exceed 4 pounds. The AC power input side of the charger shall be equipped with a line cord of a UL approved type fitted with a standard three prong wall receptacle plug. The cord shall be at least 6 ft. in length. The DC power output side of the charger shall be equipped with an electrical cable as shown on Drawing 8-1-375. The charger shall contain a double pole, double throw ON-OFF switch in the AC power line or a double pole ON-OFF switch on the front and a double pole 115 to 220 volt switch on the back, a pilot light to indicate when the charger is energized and a charging indicator light to indicate whether charger is on normal charger or trickle charger when tested as specified in 4.3.1.

3.4 Performance. The charger shall be of the constant current type capable of operating from a 115/220 volt, 47-63 Hz power source. The normal charging rate shall be 700 MA $\pm$ 10 percent when tested as specified in 4.3.1.1. The charge rate shall automatically reduce to a trickle of less than 200 MA when tested as specified in 4.3.1.2. The charger shall not be damaged when energized without a battery being connected or when deenergized with a battery connected when tested as specified in 4.3.1.

3.4.1 Burn-in. The charger shall be subjected to a burn-in period with 14 hours of continuous failure free on-time in accordance with 4.3.1.3.

3.5 Identification marking. The following shall be marked on the charger or on an attached plate in a legible and permanent manner:

Modular Battery Pack Charger for BB 541/u Batteries

Input voltage	115/220
Frequency, Hz	47-63

3.6 Labeling. The lights and switches shall be labeled to indicate their function.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:21Z
Check the source to verify that this is the current version before use.

3.7 Workmanship. The battery chargers shall be free from defects such as splits, punctures, tears, dents, creases, deteriorations or malformation. There shall be no sharp edges, slivers, burrs or projection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The end item inspection lot shall consist of all the battery chargers offered for inspection at one time. The sample unit shall be one completely assembled battery charger.

4.2.2.1 Visual examination. The charger shall be examined for the defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units.

TABLE I. List of defects

Examine	Defect
Finish	Not clean, not smooth
Construction and workmanship, general (applicable to all components and assemblies)	Any part missing or not specified type Split, punctured, torn, dented, creased, deteriorated or malformed Any sharp edge, burr or sliver
Identification markings and labels	Missing, incomplete, not legible

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:21Z
Check the source to verify that this is the current version before use.

4.2.2.2 Dimensional and weight examination. The charger shall be examined for compliance with dimensions and weight specified on Drawing 8-1-375 and 3.3. The inspection level shall be II and any dimension or weight not within specified limits shall be cause for rejection of the lot.

4.2.2.3 End item testing. Each production end item shall be tested in accordance with 4.3.1, 4.3.1.1, 4.3.1.2 and 4.3.1.3. Failure of any test shall be cause for rejection of the end item involved.

4.2.3 Packaging inspection. An examination shall be made to determine that preservation, packing and marking as required by section 5 are complied with. Defects shall be as indicated in table II. The sample unit shall be one shipping container fully packaged, except that it need not be closed. The inspection level shall be S-2 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

TABLE II. Packaging inspection

Examine	Defect
Marking (exterior and interior container)	Omitted; incorrect; not legible; of improper size, location, sequence, or method of application
Materials	Component missing, damaged, defective, or not as specified
Workmanship	Cleaning, drying and cushioning not as specified Electric cords not secured as specified Bulged or distorted container
Contents	Number per container not as specified

4.3 Methods of inspection.

4.3.1 Performance and functioning tests. Each charger shall be tested for proper functioning and for normal and trickle charging rates in accordance with 4.3.1.1 and 4.3.1.2. Failure of the current flow to be within the specified limits of 3.4 or failure of any component to operate properly during testing (such as the ON-OFF switch, pilot light and normal/trickle indicator light) shall be cause for rejection of the end item involved. Prior to being tested at normal charge rate the charger shall be energized for a minimum of 1 minute without being connected to a battery. Two batteries shall then be connected to the charger and the charger shall be disconnected from its power source for a minimum of 1 minute. Chargers which subsequently pass the tests for charge rates (4.3.1.1 and 4.3.1.2) will have thereby demonstrated conformance with the requirements of 3.4.

Source: https://assist.dla.mil -- Downloaded: 20260-21T23:21Z
Check the source to verify that this is the current version before use.

4.3.1.1 Normal charge rate test. Each charger shall be tested for the normal charge rate by energizing the charger and connecting a 19.28 ohm, 20 watt (minimum) resistor across the output terminals and measuring the current flow with an ammeter connected in series with the resistor. The ammeter shall indicate 700 MA ± 70 MA.

4.3.1.2 Trickle charge rate test. Each charger shall be tested for the trickle charge rate as specified in 4.3.1.1 except that a 72 ohm, 5 watt (minimum) resistor shall be used. The ammeter shall indicate less than 200 MA.

4.3.1.3 Burn-in. Each charger shall be subjected to a burn-in period with 14 hours of continuous failure free on-time. The test shall be conducted at room ambient temperature using an input of 115 V alternating current ± 10 percent, 60 Hz and an output of 700 MA direct current ± 10 percent. If a failure occurs, the equipment shall be repaired and returned to testing until it has accumulated 14 hours of continuous on-time without a relevant failure. All failures shall be recorded and analyzed. A relevant failure for burn-in purposes shall be defined as the inability of the equipment to meet the requirements of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Charger, battery shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Charger, battery shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Charger, battery. Each battery charger shall be individually unit packed method III as follows: Coil the cord and cable assemblies and secure to the unit with tape conforming to PPP-T-45, type III. Cushion each unit by wrapping with paperboard conforming to PPP-P-291, type III. Secure the wrap with tape conforming to PPP-T-45, type III. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, grade W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close and waterproof the box in accordance with the appendix of the box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:21Z
Check the source to verify that this is the current version before use.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of charger, battery unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization. Palletization shall be required when containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of charger, battery unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, shipping containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit pack is shipped to a single destination no packing is required; the unit container shall serve as the shipping pack. Reinforcing shall be as specified above.

Source: https://assist.dla.mil -- Downloaded: 20260-21T23:21Z
Check the source to verify that this is the current version before use.

5.2.2.1 Unitization. Palletization shall be required when unit pack quantities per destination comprise a pallet load and are not packed as specified in 5.2.2. A quantity of charger, battery unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, grade W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract, unit packs, shipping containers, and palletized loads shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract, unit packs, shipping containers, and palletized loads shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The battery chargers are for charging the batteries used with the Protective Outfit, Toxicological, Microclimate Controlled.

6.1.1 Description. The battery charger is a device capable of converting 110 volt, 60 Hz into a constant current for charging two BB 541/u batteries connected in series. The power characteristics of a BB 541/u battery are described in table III.

TABLE III. BB-541/u power characterstics

Cell type	"F" sized, sealed, nickel cadmium
Number of cells	5 cells each
Voltage/cell	1.25 volts
Nominal capacity	7.0 ampere-hours
Constant current drain rate	3 amperes

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Degree of preservation, packaging and packing required (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:21Z
Check the source to verify that this is the current version before use.

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

Custodians:

Army - GL
Air Force - 11

Review activity:

Army - AL

Preparing activity:

Army - GL

Project No. 6130-0225

☆U.S. GOVERNMENT PRINTING OFFICE: 1982-505-022/1537

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:21Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-44009A Charger, Battery, PP-7562/U

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 20261O-21T23:21Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development
Laboratories
ATTN: DRDNA-ES
Natick, MA 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:21Z
Check the source to verify that this is the current version before use.